Chebyshev Collocation Spectral Method for Three-Dimensional Transient Coupled Radiative–Conductive Heat Transfer

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A Chebyshev collocation spectral method (CSM) is presented to solve transient coupled radiative and conductive heat transfer in three-dimensional absorbing, emitting, and scattering medium in Cartesian coordinates. The walls of the enclosures are considered to be opaque, diffuse, and gray and have specified temperature boundary conditions. The CSM is adopted to solve both the radiative transfer equation (RTE) and energy conservation equation in spatial domain, and the discrete ordinates method (DOM) is used for angular discretization of RTE. The exponential convergence characteristic of the CSM for transient coupled radiative and conductive heat transfer is studied. The results using the CSM show very satisfactory calculations comparing with available results in the literature. Based on this method, the effects of various parameters, such as the scattering albedo, the conduction–radiation parameter, the wall emissivity, and the optical thickness, are analyzed. [DOI: 10.1115/1.4006596]

Keywords: three dimensional, transient, coupled radiative and conductive, Chebyshev collocation spectral method, radiative transfer equation

1 Introduction

The research of coupled radiative and conductive heat transfer within absorbing, emitting, and scattering medium plays an important role in many industry applications, such as combustion chambers, spacecraft, high temperature heat exchangers, industry furnaces, surface and porous IR-radiant burners, and porous volumetric solar receivers. As early as 1962, Viskanta and Grosh [1] developed an iterative method to analyze the one dimensional (1D) coupled radiative and conductive heat transfer. In 1998, Siegel [2] gave a detailed review for the coupled radiative and conductive heat transfer over the past 40 yr and pointed out that the transient effects of thermal radiation in participating media were important for many applications, such as glass manufacturing and ceramics at high temperature. In recent 15 years, Blobner et al. [3], Lazard et al. [4], Muresan et al. [5], David et al. [6], Asllanaj et al. [7–10], Mishra et al. [11–13], Henshall and Palmer [14], Yi et al. [15,16], Luo and Shen [17], and Chaabane [18,19] have researched the transient coupled radiative and conductive heat transfer problems.

It is well known that there are some typical numerical methods for radiative heat transfer problems, e.g., the Monte Carlo method, the zonal method, the DOM, the finite volume method (FVM), the discrete transfer method, the heat flux method, the spherical harmonics method, and the finite element method. Generally, these methods just offer h convergence (linear convergence), i.e., the convergence gained by reducing the element size h or h refinement; as a result, remeshing or refining have to be done in order to gain the wanted accuracy [20]. In this paper, we use the Chebyshev collocation spectral method, which is one kind of spectral methods, to solve transient coupled conductive and radiative heat transfer problems in three-dimensional (3D) system. One advantage of CSM is that it can provide exponential convergence [21,22]. Thus, a very high accuracy can be obtained even using a small grid number.

Spectral methods are global matching methods, in another words, the computation at any given points depends not only on information from its neighboring points but also on information from the entire domain. They have been developed rapidly in the past three decades and have been widely applied in computational fluid dynamics [23,24], quantum mechanics [25], magnetohydrodynamics [26], numerical weather prediction [27], and ocean dynamics [28]. To develop the spectral methods for radiative heat transfer, Zenouzi and Yener [29] applied the Galerkin method to solve the radiation part of the coupled radiation and convection heat transfer; Kuo et al. [30] made numerical comparisons between spectral methods and FVM in solving the radiation and natural convection combined problem, and they concluded that the spectral methods were more accurate; De Oliveira et al. [31] used the coupled Laplace transform and spectral methods to solve radiative heat transfer in isotropic scattering medium; and Zhao and Liu [32] applied the spectral element method to solve coupled radiative and conductive heat transfer in semitransparent medium. Recently, Li and co-workers have successfully developed the CSM for radiative heat transfer [33–35] and coupled radiative and conductive heat transfer [36,37].

In this paper, we extend the CSM to solve 3D transient coupled radiative and conductive heat transfer problems in an absorbing, emitting, and scattering medium. In this paper, the physical model and mathematical formulation will be presented in Sec. 2. Validations by typical cases with available numerical results are made in Sec. 3. Finally, Sec. 4 gives the conclusions.

2 Physical Model and Mathematical Formulation

A 3D cubical enclosure as shown in Fig. 1 is considered. The medium in this cube is absorbing, emitting, and scattering, and the

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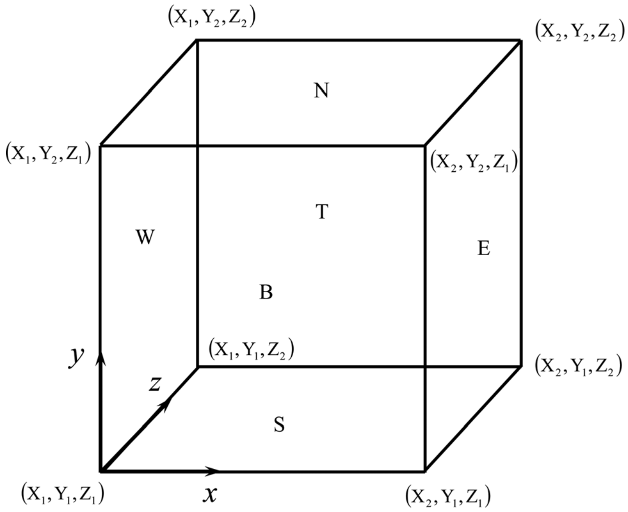

Fig. 1 Physical model and notations

properties of this medium, such as absorption coefficient κ_a, scattering coefficient κ_s, and thermal conductivity λ, are assumed to be constant. Initially, the medium is at temperature $T_g = T_N = T_W = T_E = T_B = T_T = T_{\mathrm{ref}}$, where T_{ref} is the reference temperature. For time $t > 0$, the south boundary temperature is raised to $T_S = 2T_{\mathrm{ref}}$.

For 3D transient coupled radiative and conductive heat transfer, the transient radiative term is neglected because the radiative propagation is much faster than the thermal response of the medium, then the dimensionless RTE can be written as

$$\frac{1}{\tau_{\mathrm{L}}}(\mathbf{\Omega}\cdot\nabla)G(\mathbf{r}^*,\mathbf{\Omega}) + G(\mathbf{r}^*,\mathbf{\Omega}) = (1-\omega)\Theta^4(\mathbf{r}^*) + \frac{\omega}{4\pi}\int_{4\pi} G(\mathbf{r}^*,\mathbf{\Omega}')\Phi(\mathbf{\Omega},\mathbf{\Omega}')d\mathbf{\Omega}' \tag{1}$$

For a gray, opaque, and diffuse boundary, the dimensionless radiative boundary condition for Eq. (1) can be expressed as

$$G\left(\mathbf{r}_w^*,\mathbf{\Omega}\right) = \varepsilon_w\Theta^4\left(\mathbf{r}_w^*\right) + \frac{1-\varepsilon_w}{\pi}\int_{\mathbf{n}_\omega\cdot\mathbf{\Omega}'>0} G\left(\mathbf{r}_w^*,\mathbf{\Omega}'\right)\Phi(\mathbf{\Omega}',\mathbf{\Omega})d\mathbf{\Omega}', \quad \mathbf{n}_w\cdot\mathbf{\Omega} < 0 \tag{2}$$

where the optical thickness τ_{L}, the dimensionless radiative intensity G, the dimensionless spatial coordinates vector $\mathbf{r}^*$, the dimensionless temperature Θ, and the scattering albedo ω are defined as follows:

$$\tau_{\mathrm{L}} = \beta L, \quad G = \frac{\pi I}{\sigma T_{\mathrm{ref}}^4}, \quad \mathbf{r}^* = \mathbf{r}/L, \quad \Theta = T/T_{\mathrm{ref}}, \quad \omega = \kappa_s/\beta \tag{3}$$

The discrete ordinates equation is obtained by evaluating Eq. (1) in each discrete direction and replacing the integral by a numerical quadrature

$$\frac{1}{\tau_{\mathrm{L}}}(\mathbf{\Omega}^m\cdot\nabla)G(\mathbf{r}^*,\mathbf{\Omega}^m) + G(\mathbf{r}^*,\mathbf{\Omega}^m) = (1-\omega)\Theta^4(\mathbf{r}^*) + \frac{\omega}{4\pi}\sum_{m'=1}^{M} G\left(\mathbf{r}^*,\mathbf{\Omega}^{m'}\right)\Phi^{m,m'}w^{m'} \tag{4}$$

Similarly, the discrete ordinates form of the boundary condition, Eq. (2), can be expressed as

$$G\left(\mathbf{r}_w^*,\mathbf{\Omega}^m\right) = \varepsilon_w\Theta^4\left(\mathbf{r}_w^*\right) + \frac{1-\varepsilon_w}{\pi}\sum_{\substack{m'=1,\\ n_w\cdot\Omega^{m'}>0}}^{M} G\left(\mathbf{r}_w^*,\mathbf{\Omega}^{m'}\right)\Phi\left(\mathbf{\Omega}^{m'},\mathbf{\Omega}^m\right)w^{m'}, \quad \mathbf{n}_w\cdot\mathbf{\Omega}^m < 0 \tag{5}$$

where $\Omega^m = i\mu^m + j\eta^m + k\xi^m$.

The energy conservation equation of transient coupled radiative and conductive heat transfer is given as

$$\frac{\partial\Theta}{\partial t^*} = \frac{1}{\tau_{\mathrm{L}}^2}\nabla^2\Theta - \frac{1-\omega}{N_{cr}}\left[\Theta^4 - \frac{1}{4\pi}\int_{4\pi} G(\mathbf{r}^*,\mathbf{\Omega},t^*)d\mathbf{\Omega}\right] \tag{6}$$

with the initial and boundary conditions

$$\Theta(\mathbf{r}^*,0) = \text{known} \tag{7}$$

$$\Theta\left(\mathbf{r}_w^*,t^*\right) = \text{known} \tag{8}$$

where $t^* = \frac{\lambda\beta^2 t}{\rho c_p}$ and $N_{cr} = \frac{\lambda\beta}{4\sigma T_{\mathrm{ref}}^3}$ represent the dimensionless time and the conduction–radiation parameter, respectively.

Using the general Euler forward scheme, the temporal discretization of the energy conservation equation is obtained as

$$\begin{aligned}\frac{\Theta^{n+1}-\Theta^n}{\Delta t^*} = f\Bigg\{&\frac{1}{\tau_{\mathrm{L}}^2}\nabla^2\Theta^{n+1} - \frac{1-\omega}{N_{cr}} \\ &\times\left[\left(\Theta^{n+1}\right)^4 - \frac{1}{4\pi}\int_{4\pi} G^{n+1}d\mathbf{\Omega}\right]\Bigg\} + (1-f) \\ &\times\left\{\frac{1}{\tau_{\mathrm{L}}^2}\nabla^2\Theta^n - \frac{1-\omega}{N_{cr}}\left[\left(\Theta^n\right)^4 - \frac{1}{4\pi}\int_{4\pi} G^n d\mathbf{\Omega}\right]\right\}\end{aligned} \tag{9}$$

where Δt^* is the dimensionless time step and $f\in[0,1]$ is an adjustment parameter. Taking f as $1/2$, $2/3$, and 1, the Crank–-Nicolson, Galerkin, and fully implicit schemes are obtained, respectively.

For both dimensionless RTE (Eq. (4)) and dimensionless energy conservation equation (Eq. (9)), the spatial domain is discretized by the Chebyshev–Gauss–Lobatto collocation points

$$\begin{cases} r_{x,i} = -\cos\dfrac{\pi i}{N_x}, & i = 0,\ 1,\ \cdots,\ N_x \\ r_{y,j} = -\cos\dfrac{\pi j}{N_y}, & j = 0,\ 1,\ \cdots,\ N_y \\ r_{z,k} = -\cos\dfrac{\pi k}{N_z}, & i = 0,\ 1,\ \cdots,\ N_z \end{cases} \tag{10}$$

where N_x, N_y, and N_z are the Chebyshev polynomial nodes in x,y, and z directions, respectively.

To meet the requirements of the Chebyshev–Gauss–Lobatto collocation points, the arbitrary intervals $[x_{\min},x_{\max}]\times[y_{\min},y_{\max}]\times[z_{\min},z_{\max}]$ need to be mapped to standard intervals $[-1,1]\times[-1,1]\times[-1,1]$

$$\begin{cases} x = \dfrac{r_x(x_{\max}-x_{\min}) + (x_{\min}+x_{\max})}{2} \\ y = \dfrac{r_y(y_{\max}-y_{\min}) + (y_{\min}+y_{\max})}{2} \\ z = \dfrac{r_z(z_{\max}-z_{\min}) + (z_{\min}+z_{\max})}{2} \end{cases} \tag{11}$$

The dimensionless temperature and the dimensionless radiative intensity can be approximated by the Lagrange interpolation polynomial, such as

$$\begin{cases} \Theta(x,y,z) = \sum_{i=0}^{N_x}\sum_{j=0}^{N_y}\sum_{k=0}^{N_z}\Theta(x_i,y_j,z_k)h_i(x)h_j(y)h_k(z) \\ G^m(x,y,z) = \sum_{i=0}^{N_x}\sum_{j=0}^{N_y}\sum_{k=0}^{N_z}G^m(x_i,y_j,z_k)h_i(x)h_j(y)h_k(z) \end{cases} \tag{12}$$

where $h_i(x)$, $h_j(y)$, and $h_k(z)$ are the functions of the first kind Chebyshev polynomials.

Substituting Eq. (12) into Eq. (9), we can obtain the discretized algebraic equation

$$\sum_{l=0}^{N_x} P_{il}\Theta_{ljk} + \sum_{l=0}^{N_y} Q_{jl}\Theta_{ilk} + \sum_{l=0}^{N_z} R_{kl}\Theta_{ijl} = V_{ijk} \tag{13}$$

where the element expressions for matrices **P**, **Q**, **R**, and **V** are

$$P_{il} = \begin{cases} \dfrac{4f\Delta t^*}{\tau_L^2(x_{\max}-x_{\min})^2}D_{il}^{(2)} - \dfrac{1}{3}, & i=l \\ \dfrac{4f\Delta t^*}{\tau_L^2(x_{\max}-x_{\min})^2}D_{il}^{(2)}, & i\neq l \end{cases} \tag{14}$$

$$Q_{jl} = \begin{cases} \dfrac{4f\Delta t^*}{\tau_L^2(y_{\max}-y_{\min})^2}D_{jl}^{(2)} - \dfrac{1}{3}, & j=l \\ \dfrac{4f\Delta t^*}{\tau_L^2(y_{\max}-y_{\min})^2}D_{jl}^{(2)}, & j\neq l \end{cases} \tag{15}$$

$$R_{kl} = \begin{cases} \dfrac{4f\Delta t^*}{\tau_L^2(z_{\max}-z_{\min})^2}D_{kl}^{(2)} - \dfrac{1}{3}, & k=l \\ \dfrac{4f\Delta t^*}{\tau_L^2(z_{\max}-z_{\min})^2}D_{kl}^{(2)}, & k\neq l \end{cases} \tag{16}$$

$$\begin{aligned} V_{ijk} = {} & \frac{f\Delta t^*}{N_{cr}}(1-\omega)\left[\left(\Theta_{ijk}^{n+1,\text{previous}}\right)^4 + \frac{1}{4\pi}\sum_{m=1}^{M}G_{ijk}^{m,n+1}w^m\right] - \Theta_{ijk}^n \\ & - \frac{\Delta t^*}{\tau_L^2}(1-f)\left(\sum_{l=0}^{N_x}D_{il}^{(2)}\Theta_{ljk}^n + \sum_{l=0}^{N_y}D_{jl}^{(2)}\Theta_{ilk}^n + \sum_{k=0}^{N_z}D_{kl}^{(2)}\Theta_{ijl}^n\right) \\ & + \frac{\Delta t^*}{N_{cr}}(1-f)(1-\omega)\left[\left(\Theta_{ijk}^n\right)^4 + \frac{1}{4\pi}\sum_{m=1}^{M}G_{ijk}^{m,n}w^m\right] \end{aligned} \tag{17}$$

where $\mathbf{D}^{(2)}$ is the second order derivative matrix corresponding to the Chebyshev–Gauss–Lobatto points, and the elements can be expressed as Eq. (18) [21,23]. The superscript *previous* denotes the value of former-computation in this time level

$$D_{il}^{(2)} = \begin{cases} \dfrac{(-1)^{i+l}}{c_l}\dfrac{r_i^2+r_ir_l-2}{(1-r_i^2)(r_i-r_l)^2}, & 1\le i\le N-1, 0\le l\le N, i\neq l \\ -\dfrac{(N^2-1)(1-r_i^2)+3}{3(1-r_i^2)^2}, & 1\le i=l\le N-1 \\ \dfrac{2}{3}\dfrac{(-1)^l}{c_l}\dfrac{(2N^2+1)(1-r_l)-6}{(1-r_l)^2}, & i=0, 1\le l\le N \\ \dfrac{2}{3}\dfrac{(-1)^{(l+N)}}{c_l}\dfrac{(2N^2+1)(1-r_l)-6}{(1+r_l)^2}, & i=N, 1\le l\le N-1 \\ \dfrac{N^4-1}{15}, & i=l=0, i=l=N \end{cases} \tag{18}$$

Similarly, the spectral discretization of the dimensionless RTE can be written in matrix form

$$\sum_{l=0}^{N_x} A_{il}^m G_{ljk}^m + \sum_{l=0}^{N_y} B_{jl}^m G_{ilk}^m + \sum_{l=0}^{N_z} C_{kl}^m G_{ijl}^m = F_{ijk}^m \tag{19}$$

where the element expressions for matrices $\mathbf{A}^m$, $\mathbf{B}^m$, $\mathbf{C}^m$, and $\mathbf{F}^m$ are

$$A_{il}^m = \begin{cases} \dfrac{2\mu^m}{\tau_\mathrm{L}(x_{\max}-x_{\min})}D_{il}^{(1)} + \dfrac{1}{3}, & i=l \\ \dfrac{2\mu^m}{\tau_\mathrm{L}(x_{\max}-x_{\min})}D_{il}^{(1)}, & i\neq l \end{cases} \tag{20}$$

$$B_{jl}^m = \begin{cases} \dfrac{2\eta^m}{\tau_\mathrm{L}(y_{\max}-y_{\min})}D_{jl}^{(1)} + \dfrac{1}{3}, & j=l \\ \dfrac{2\eta^m}{\tau_\mathrm{L}(y_{\max}-y_{\min})}D_{jl}^{(1)}, & j\neq l \end{cases} \tag{21}$$

$$C_{kl}^m = \begin{cases} \dfrac{2\xi^m}{\tau_\mathrm{L}(z_{\max}-z_{\min})}D_{kl}^{(1)} + \dfrac{1}{3}, & k=l \\ \dfrac{2\xi^m}{\tau_\mathrm{L}(z_{\max}-z_{\min})}D_{kl}^{(1)}, & k\neq l \end{cases} \tag{22}$$

$$F_{ijk}^m = (1-\omega)\Theta_{ijk}^4 + \frac{\omega}{4\pi}\sum_{m'=1}^{M}G_{ijk}^{m',\text{previous}}\Phi^{m',m}w^{m'} \tag{23}$$

where $\mathbf{D}^{(1)}$ is the first order derivative matrix corresponding to the Chebyshev–Gauss–Lobatto points, and the elements can be expressed as Eq. (24) [21,23]

$$D_{il}^{(1)} = \begin{cases} \dfrac{c_i}{c_l}\dfrac{(-1)^{i+l}}{r_i-r_l}, & j\neq l \\ -\dfrac{r_l}{2(1-r_l^2)} & 1\le j=l\le N-1 \\ \dfrac{2N^2+1}{6}, & j=l=0 \\ -\dfrac{2N^2+1}{6}, & j=l=N \end{cases} \tag{24}$$

The implementations of the CSM for solving 3D transient coupled radiative and conductive heat transfer problems can be carried out according to the following routines:

Step 1. Choose the number of collocation points, N_x, N_y, and N_z, compute the coordinate values of nodes, and give the dimensionless radiative intensity and the dimensionless temperature initial assumptions (zero, for example, except for boundaries).

Step 2. Compute matrices $\mathbf{D}^{(1)}$, $\mathbf{D}^{(2)}$, **A**, **B**, **C**, **P**, **Q**, and **R** once for all.

Step 3. Loop at each time step, $n = 1, 2, \cdots, N_t$.

Step 4. Assemble the matrix V (Eq. (17)) and impose the initial condition (Eq. (7)) and the Dirichlet boundary condition (Eq. (8)); then directly solve Eq. (13) to obtain the updated dimensionless temperature $\Theta^{n+1,\text{new}}$, where the superscript *new* denotes the most updated new value in this time level. Make under-relaxation to the temperature field as needed.

Step 5. Assemble the matrix F^m (Eq. (23)) and impose the boundary condition (Eq. (5)); then directly solve Eq. (19) to obtain the dimensionless radiative intensities $G^{m,\text{new}}$.

Step 6. If the convergence criteria (Eqs. (25 and 26)) are satisfied, then terminate the iterations and go to step 7. Otherwise, go back to step 4 and use the new $\Theta^{n+1,\text{new}}$ and $G^{m,\text{new}}$ instead of the previous $\Theta^{n+1,\text{previous}}$ and $G^{m,\text{previous}}$.

Step 7. If the maximum time has not been reached, go to step 3, else do postprocessing.

In above routines, the most important steps are solving the matrix form of energy conservation equation and discretized RTE. Energy conservation equation can be directly solved by two-steps direct solver [38] and RTE can be directly solved by 3D Schur-decomposition [35].

The convergence criteria of dimensionless temperature and dimensionless radiative intensity are

$$\frac{\left\|\Theta^{n+1,\text{new}} - \Theta^{n+1,\text{previous}}\right\|}{\left\|\Theta^{n+1,\text{new}}\right\|} \leq 10^{-6} \tag{25}$$

$$\frac{\left\|G^{m,\text{new}} - G^{m,\text{previous}}\right\|}{\left\|G^{m,\text{new}}\right\|} \leq 10^{-6} \tag{26}$$

3 Results and Discussions

Considering a 3D cubical enclosure of semitransparent participating medium, the effects of scattering albedo, conduction–-radiation parameter, wall emissivity, and optical thickness on transient coupled radiative and conductive heat transfer are analyzed. Before these analyses, we first examined the convergence characteristic of the CSM for transient coupled radiative and conductive heat transfer in different spatial nodes.

For transient heat transfer, the dimensionless time step $\Delta t^* = 1.0 \times 10^{-4}$ is used, and the steady-state conduction was assumed to have been achieved when the maximum absolute difference of Θ at any location between two consecutive time steps did not exceed 1.0×10^{-6}. For the sake of quantitative analysis and comparison of the CSM results ($R_{\text{CSM}}(x)$) with the benchmark results ($R_{\text{Benchmark}}(x)$), the integral relative error is defined as

$$\text{Integral relative error\%} = \frac{\int |R_{\text{CSM}}(x) - R_{\text{Benchmark}}(x)| dx}{\int |R_{\text{Benchmark}}(x)| dx} \times 100\% \tag{27}$$

For the case of $N_{cr} = 0.01$, $\tau_L = 1.0$, $\omega = 0.5$, and $\varepsilon = 1.0$, the integral relative errors are given in Fig. 2, where the numerical results obtained using $N_x \times N_y \times N_z = 28 \times 28 \times 28$ are considered as the benchmark results. It can be seen that the convergence rate of the CSM is very fast and approximately follows an exponential law trend.

Then, we investigate the effects of nodes and angular directions on dimensionless temperature by comparing the transient state results at dimensionless time $t^* = 0.005$ along the centerline

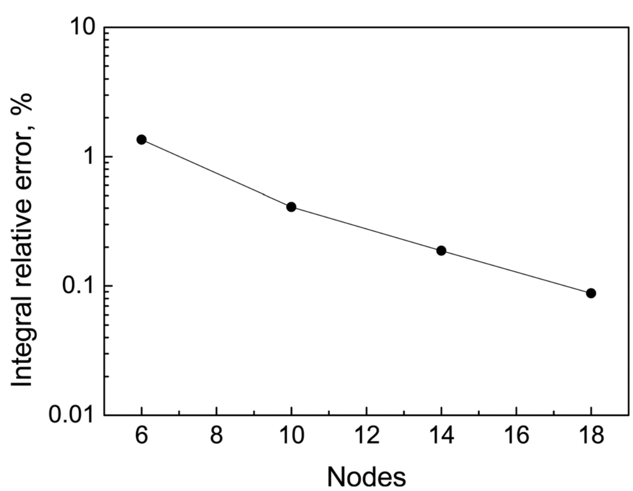

Fig. 2 The exponential convergence rate of CSM for the transient coupled radiation–conduction heat transfer according to the total number of spatial nodes

Table 1 Effects of nodes and angular directions by CSM on dimensionless temperature at three locations for $N_{cr} = 0.01$, $\tau_L = 1.0$, $\omega = 0.0$, $\varepsilon = 1.0$, and $t^* = 0.005$

Nodes	Angular directions	$y^* = 0.25$	$y^* = 0.50$	$y^* = 0.75$
Effect of nodes				
$6 \times 6 \times 6$	S_8	0.59771 (0.234%)	0.53896 (0.348%)	0.51587 (0.068%)
$10 \times 10 \times 10$	S_8	0.59785 (0.210%)	0.54073 (0.020%)	0.51585 (0.072%)
$14 \times 14 \times 14$	S_8	0.59916 (0.008%)	0.54097 (0.024%)	0.51635 (0.025%)
$18 \times 18 \times 18$	S_8	**0.59911**	**0.54084**	**0.51622**
Effect of angular directions				
$14 \times 14 \times 14$	S_4	0.60339 (0.888%)	0.5444 (0.766%)	0.51798 (0.390%)
$14 \times 14 \times 14$	S_6	0.60089 (0.470%)	0.54232 (0.381%)	0.51682 (0.165%)
$14 \times 14 \times 14$	S_8	0.59916 (0.181%)	0.54097 (0.131%)	0.51635 (0.074%)
$14 \times 14 \times 14$	S_{10}	**0.59808**	**0.54026**	**0.51597**

($x^*, y^*, z^* = 0.5$). The results that are obtained for $N_{cr} = 0.01$, $\tau_L = 1.0$, $\omega = 0.0$, and $\varepsilon = 1.0$ are listed in Table 1. It can be seen that when the nodes are $14 \times 14 \times 14$, the maximum variation in dimensionless temperature between two finest nodes level $14 \times 14 \times 14$ and $18 \times 18 \times 18$ is less than 0.025% and as such no further refinement is carried out. With $14 \times 14 \times 14$ nodes, the results of the different orders of S_N approximation on dimensionless temperature are also listed in Table 1. No significant differences can be found for S_8 and S_{10} approximations. The trend observed with other sets of parameters is similar. Therefore, in the following, we use $14 \times 14 \times 14$ nodes for space discretization and S_8 approximation for angular direction.

Figure 3 highlights contour plot of transient isotherms at center plane ($x^* = 0.5$, y^*, z^*) for $t^* = 0.005$, $t^* = 0.015$, $t^* = 0.050$, and $t^* = SS$. In this case, the input parameters are $\tau_L = 1.0$, $\omega = 0$, $N_{cr} = 0.01$, and $\varepsilon = 1$.

In Fig. 4, the dimensionless temperature Θ by using the CSM and the combined lattice Boltzmann method and finite volume method (LBM-FVM) [11] on centerline ($x^* = 0.5$, y^*, $z^* = 0.5$) are compared for different ω. With $\tau_L = 1.0$, $\varepsilon = 1.0$, and $N_{cr} = 0.01$, these comparisons are shown in Fig. 4(*a*) and Fig. 4(*b*) for $\omega = 0.5$ and $\omega = 0.9$, respectively. It is seen that for both $\omega = 0.5$ and $\omega = 0.9$, the results by using the CSM are in good agreement with those of LBM-FVM. For $\omega = 0.5$, along with dimensionless time $t^* = 0.005$, $t^* = 0.015$, and $t^* = SS$, the maximum relative errors of the CSM results are 1.439%, 2.452%, and 1.961%, respectively, compared with the values of LBM-FVM [11]. Similarly, for $\omega = 0.9$, the maximum relative errors of CSM and LBM-FVM are 2.603%, 3.406%, 1.981%, and 1.102% for $t^* = 0.005$, $t^* = 0.015$, $t^* = 0.050$, and $t^* = SS$.

Figure 5 shows Θ at different t^* on centerline ($x^* = 0.5$, y^*, $z^* = 0.5$) for two different conduction–radiation parameters, namely, $N_{cr} = 0.1$ and $N_{cr} = 1.0$. It is seen that for both radiation dominated ($N_{cr} = 0.1$) and conduction dominated ($N_{cr} = 1.0$) cases, the results of CSM agree with those of the LBM-FVM [11] very well. In Figs. 5(*a*) and 5(*b*), at early stages $t^* = 0.005$ and $t^* = 0.015$, it seems that the CSM results are some distance ahead of the LBM-FVM results from the hot boundary. The reason is that, for different methods, the convergence rates are different, transient results may differ in some disparity, but the differences are small, the maximum relative error of these two methods is less than 4.213%, and the maximum relative errors of the steady-state Θ by two methods are 1.132% and 0.191%, respectively.

For $\tau_L = 1.0$, $N_{cr} = 0.01$, and $\omega = 0$, the CSM approximation of Θ at different t^* on centerline ($x^* = 0.5$, y^*, $z^* = 0.5$) for two wall emissivities of south boundary $\varepsilon_s = 0.1$ (*a*) and $\varepsilon_s = 0.5$ (*b*) are shown in Fig. 6. Compared to these results obtained by

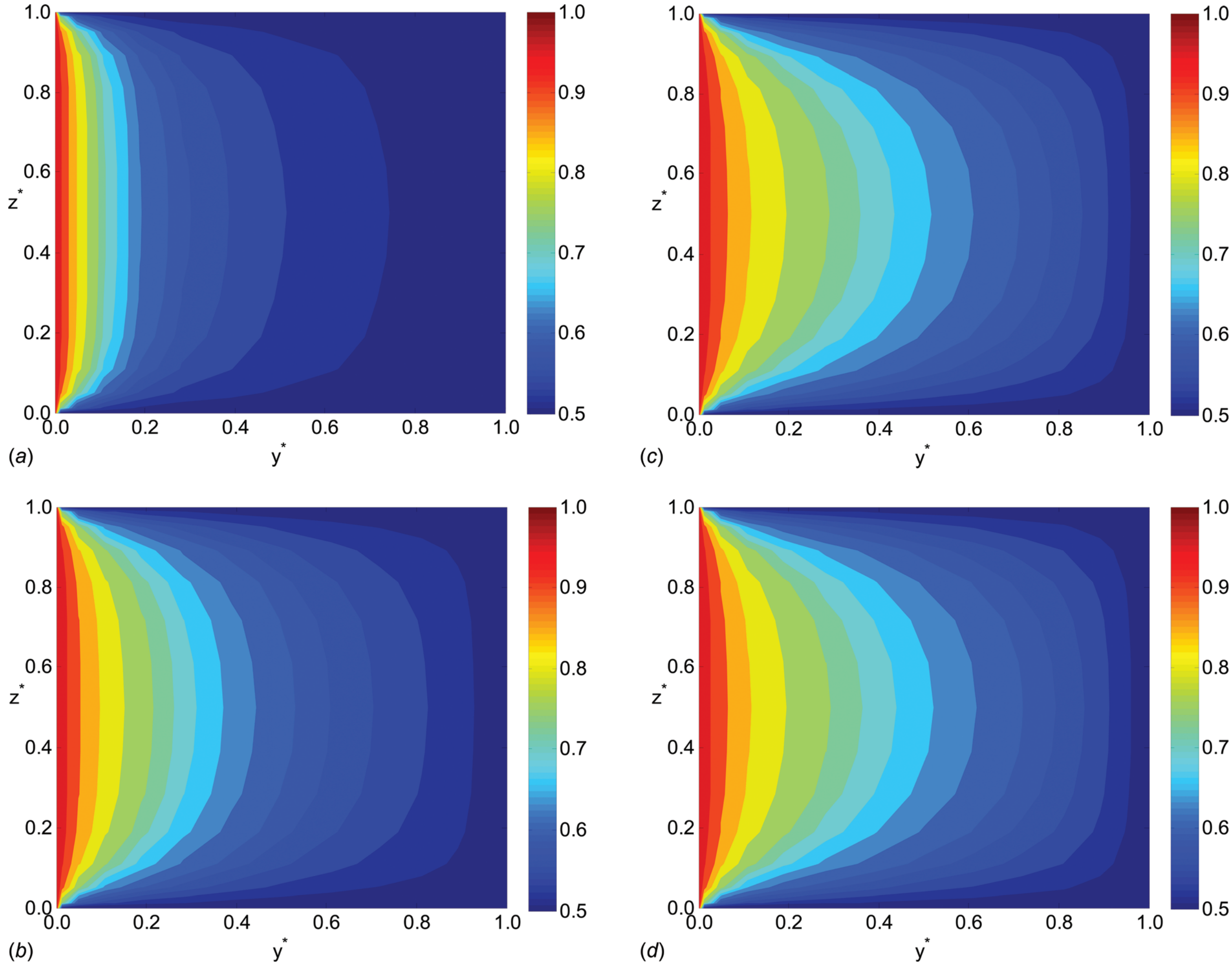

Fig. 3 For different dimensionless time t^*, isotherms at the center plane ($x^* = 0.5$, y^*, z^*) for the case of $\tau_L = 1.0$, $\omega = 0$, $\varepsilon = 1$, and $N_{cr} = 0.01$. (*a*) $t^* = 0.005$; (*b*) $t^* = 0.015$; (*c*) $t^* = 0.050$; and (*d*) $t^* = SS$

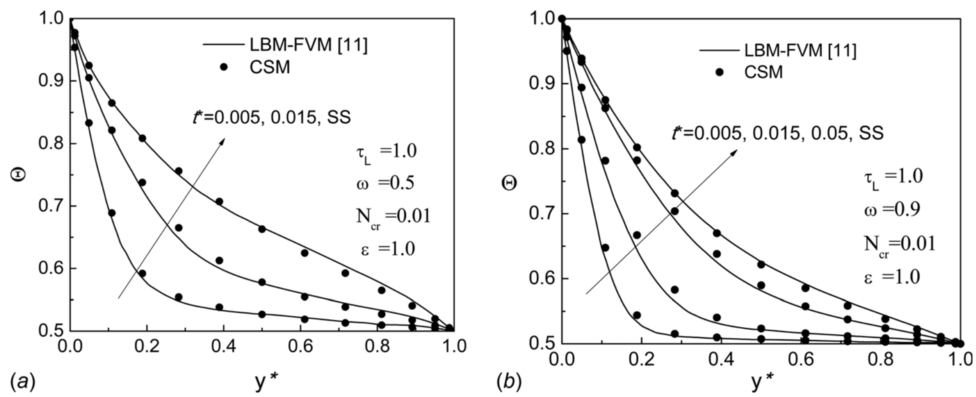

Fig. 4 For $\tau_L = 1.0$, $\varepsilon = 1$, and $N_{cr} = 0.01$, dimensionless temperature on the centerline ($x^* = 0.5, y^*$, $z^* = 0.5$) with different t^*. (*a*) $\omega = 0.5$ and (*b*) $\omega = 0.9$

Mondal and Mishra [11], the results by the CSM are very close to those by the LBM-FVM. For $\varepsilon_s = 0.1$, the maximum relative errors of the these two methods are 1.620%, 1.169%, and 0.608% with different $t^* = 0.005$, $t^* = 0.015$, and $t^* = SS$. For $\varepsilon_s = 0.5$, the maximum relative errors of the CSM results are 1.053%, 0.801%, and 0.972%, respectively.

For $\omega = 0$, $\varepsilon = 1$, and $N_{cr} = 0.01$, Fig. 7 shows Θ at different t^* on centerline ($x^* = 0.5$, y^*, $z^* = 0.5$) for two different conditions of the optical thickness, namely, $\tau_L = 0.1$ and $\tau_L = 5.0$. As shown in Fig. 7, the CSM results are in good agreement with the results obtained by the LBM-FVM [11]. It is noted that, the level of nodes $N_x \times N_y \times N_z = 14 \times 14 \times 14$ is used for the CSM; however, $N_x \times N_y \times N_z = 31 \times 31 \times 31$ control volumes/lattices are used for the LBM-FVM [11]. These results show that, even using a small number of nodes, the present method can provide high accuracy to solve this problem.

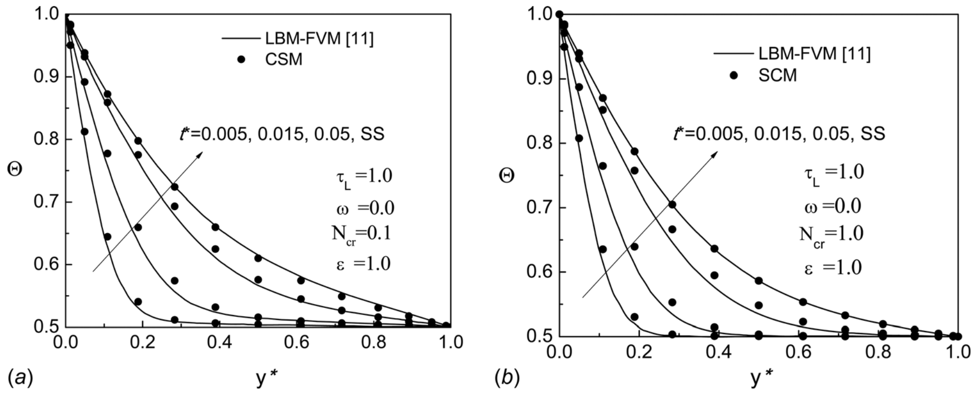

Fig. 5 For $\tau_L = 1.0$, $\varepsilon = 1$, and $\omega = 0$, dimensionless temperature on the centerline ($x^* = 0.5$, y^*, $z^* = 0.5$) with different t^*. (*a*) $N_{cr} = 0.1$ and (*b*) $N_{cr} = 1.0$

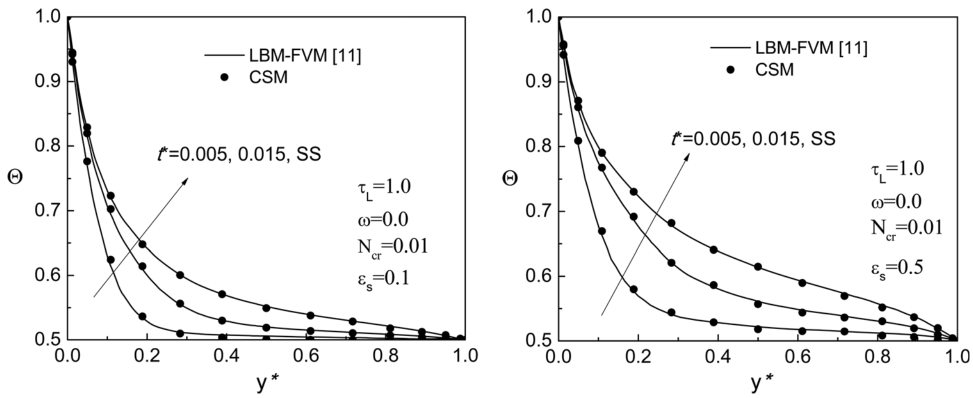

Fig. 6 For $\tau_L = 1.0$, $N_{cr} = 0.01$, and $\omega = 0$, dimensionless temperature on the centerline ($x^* = 0.5$, y^*, $z^* = 0.5$) with different t^*. (*a*) $\varepsilon_S = 0.1$ and (*b*) $\varepsilon_S = 0.5$

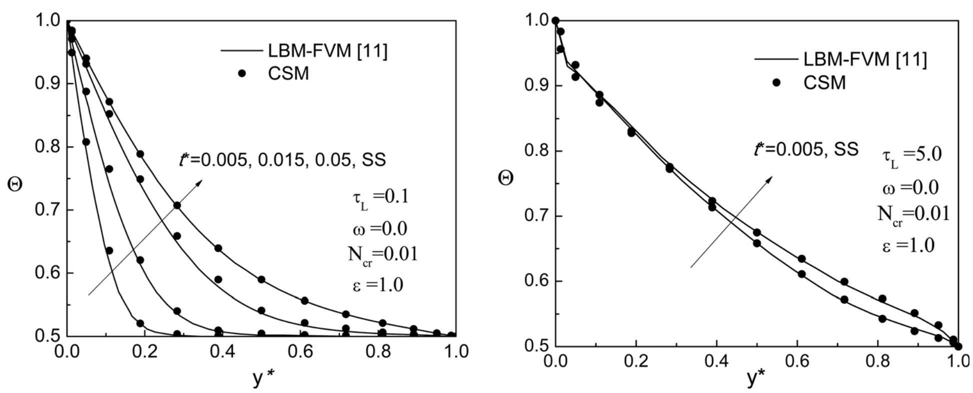

Fig. 7 For $\omega = 0$, $\varepsilon = 1$, and $N_{cr} = 0.01$, dimensionless temperature on the centerline ($x^* = 0.5$, y^*, $z^* = 0.5$) with different t^*. (*a*) $\tau_L = 0.1$ and (*b*) $\tau_L = 5.0$

4 Conclusions

The highlight of the CSM is its exponential convergence characteristic. In this paper, the transient energy conservation equation and radiative transfer equation are both solved by the CSM on the same node system to predict coupled radiative and conductive heat transfer within an absorbing, emitting, and isotropically scattering participating medium. To our knowledge, this method is applied for the first time to three-dimensional transient radiative and conductive heat transfer. Four various test cases are taken as examples to verify the performance of the presented method for various values of the scattering albedo, the conduction–radiation parameter, the wall emissivity, and the optical thickness. The predicted dimensionless temperature distributions agree well with those solutions in references. The CSM is very effective to solve transient in three-dimensional coupled radiative and conductive heat transfer in semitransparent medium even using a small number of grid.

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Nomenclature

A = matrix defined in Eq. (20)
B = matrix defined in Eq. (21)

$\mathbf{C}$ = matrix defined in Eq. (22)
c_p = specific heat of medium, $\mathrm{J \cdot kg^{-1} \cdot K^{-1}}$
$\mathbf{D}^{(1)}$ = the first order derivative matrix
$\mathbf{D}^{(2)}$ = the second order derivative matrix
$\mathbf{F}$ = matrix defined in Eq. (23)
f = time adjustment parameter
G = dimensionless radiative intensity
h = function of the first order derivative of Chebyshev polynomials
I = radiative intensity, $\mathrm{W \cdot m^{-2} \cdot sr^{-1}}$
i = solution node index for x-direction
j = solution node index for y-direction
k = solution node index for z-direction
L = characteristic length, m
M = number of discrete ordinates directions
N = total number of spatial solution nodes
N_{cr} = conduction–radiation parameter, $\lambda(\kappa_a + \kappa_s)/4\sigma T_{\text{ref}}^3$
N_t = number of discretized time steps
N_x = Chebyshev polynomial nodes in x-direction
N_y = Chebyshev polynomial nodes in y-direction
N_z = Chebyshev polynomial nodes in z-direction
$\mathbf{n}_w$ = unit outward normal vector of boundary
$\mathbf{P}$ = matrix defined in Eq. (14)
$\mathbf{Q}$ = matrix defined in Eq. (15)
$\mathbf{R}$ = matrix defined in Eq. (16)
R_{CSM} = collocation spectral method results
$R_{\text{Benchmark}}$ = benchmark results
$\mathbf{r}$ = spatial coordinates vector, m
r^* = dimensionless spatial coordinates vector, $\mathbf{r}/L$
T = temperature, K
T_{ref} = reference temperature, K
t = time, s
t^* = dimensionless time, $(\lambda\beta^2 t)/\rho c_p$
Δt^* = dimensionless time step
$\mathbf{V}$ = matrix defined in Eq. (17)
w = weight of discrete ordinates approximation, sr
x = coordinate in x-direction, m
x^* = dimensionless coordinate in x-direction, x/L
y = coordinate in y-direction, m
y^* = dimensionless coordinate in x-direction, y/L
z = coordinate in z-direction, m
z^* = dimensionless coordinate in x-direction, z/L

Greek Symbols

β = extinction coefficient, $\mathrm{m^{-1}}$
ε = emissivity
η = direction cosine in y-direction
Θ = dimensionless temperature
κ_a = absorption coefficient, $\mathrm{m^{-1}}$
κ_s = scattering coefficient, $\mathrm{m^{-1}}$
λ = thermal conductivity, $\mathrm{W \cdot m^{-1} \cdot K^{-1}}$
μ = direction cosine in x-direction
ξ = direction cosine in z-direction
ρ = density of medium, $\mathrm{kg \cdot m^{-3}}$
σ = Stefan–Boltzmann constant, $\mathrm{W \cdot m^{-2} \cdot K^{-4}}$
ω = scattering albedo
τ_L = optical thickness
Φ = scattering phase function
Ω, Ω' = unit vector of radiation direction

Subscripts

E, W, N, S = east, west, north, and south boundaries
g = value at gas medium
i, j, k = solution node indexes for x, y, z-directions
max = maximum value
min = minimum value
w = value at wall

Superscripts

m, m' = angular direction of radiation
n = time step
previous = the last iterative value

References

[1] Viskanta, R., and Grosh, R. J., 1962, "Heat Transfer by Simultaneous Conduction and Radiation in an Absorbing Medium," ASME Trans. J. Heat Transfer, **84**(1), pp. 63–72.
[2] Siegel, R., 1998, "Transient Thermal Effects of Radiant Energy in Translucent Materials," ASME Trans J. Heat Transfer, **120**(1), pp. 4–23.
[3] Blobner, J., BiaŁecki, R. A., and Kuhn, G., 1999, "Transient Non-Linear Heat Conduction–Radiation Problems—A Boundary Element Formulation," Int. J. Numer. Methods Eng., **46**(11), pp. 1865–1882.
[4] Lazard, M., André, S., and Maillet, D., 2001, "Transient Coupled Radiative-Conductive Heat Transfer in a Gray Planar Medium With Anisotropic Scattering," J. Quant. Spectrosc. Radiat. Transf., **69**(1), pp. 23–33.
[5] Muresan, C., Vaillon, R., Menezo, C., and Morlot, R., 2004, "Discrete Ordinates Solution of Coupled Conductive Radiative Heat Transfer in a Two-Layer Slab With Fresnel Interfaces Subject to Diffuse and Obliquely Collimated Irradiation," J. Quant. Spectrosc. Radiat. Transf., **84**(4), pp. 551–562.
[6] David, L., Nacer, B., Pascal, B., and Gérard, J., 2006, "Transient Radiative and Conductive Heat Transfer in Non-Gray Semitransparent Two-Dimensional Media With Mixed Boundary Conditions," Heat Mass Transfer, **42**(4), pp. 322–337.
[7] Asllanaj, F., Brige, X., and Jeandel, G., 2007, "Transient Combined Radiation and Conduction in a One-Dimensional Non-Gray Participating Medium With Anisotropic Optical Properties Subjected to Radiative Flux at the Boundaries," J. Quant. Spectrosc. Radiat. Transf., **107**(1), pp. 17–29.
[8] Asllanaj, F., Feldheim, V., and Lybaert, P., 2007, "Solution of Radiative Heat Transfer in 2-D Geometries by a Modified Finite-Volume Method Based on a Cell Vertex Scheme Using Unstructured Triangular Meshes," Numer. Heat Transfer, Part B, **51**(2), pp. 97–119.
[9] Asllanaj, F., Parent, G., and Jeandel, G., 2007, "Transient Radiation and Conduction Heat Transfer in a Gray Absorbing-Emitting Medium Applied on Two-Dimensional Complex-Shaped Domains," Numer. Heat Transfer, Part B, **52**(2), pp. 179–200.
[10] Kaemmerlen, A., Asllanaj, F., Sallée, H., Baillis, D., and Jeandel, G., 2010, "Transient Modeling of Combined Conduction and Radiation in Wood Fibre Insulation and Comparison With Experimental Data," Int. J. Therm. Sci., **49**(11), pp. 2169–2176.
[11] Mondal, B., and Mishra, S. C., 2009, "Analysis of 3-D Conduction-Radiation Heat Transfer Using the Lattice Boltzmann Method," J. Thermophys. Heat Transfer, **23**(1), pp. 210–215.
[12] Das, R., Mishra, S. C., and Uppaluri, R., 2010, "Inverse Analysis Applied to Retrieval of Parameters and Reconstruction of Temperature Field in a Transient Conduction-Radiation Heat Transfer Problem Involving Mixed Boundary Conditions," Int. Commun. Heat Mass Transfer, **37**(1), pp. 52–57.
[13] Mishra, S. C., Muthukumaran, R., and Maruyama, S., 2011, "The Finite Volume Method Approach to the Collapsed Dimension Method in Analyzing Steady/Transient Radiative Transfer Problems in Participating Media," Int. Commun. Heat Mass Transfer, **38**(3), pp. 291–297.
[14] Henshall, P., and Palmer, P., 2008, "A Leapfrog Algorithm for Coupled Conductive and Radiative Transient Heat Transfer in Participating Media," Int. J. Therm. Sci., **47**(4), pp. 388–398.
[15] Yi, H. L., Xie, M., and Tan, H. P., 2008, "Transient Coupled Heat Transfer in an Anisotropic Scattering Composite Slab With Semitransparent Surfaces," Int. J. Heat Mass Transfer, **51**(25–26), pp. 5918–5930.
[16] Yi, H. L., Zhang, H. C., and Tan, H. P., 2009, "Transient Radiation and Conduction Heat Transfer Inside a Plane-Parallel Participating Gray Medium With Boundaries Having Different Reflecting Characteristics," J. Quant. Spectrosc. Radiat. Transf., **110**(18), pp. 1978–1992.
[17] Luo, J. F., and Shen, X., 2009, "Numerical Method of the Ray Tracing-Node Analyzing Method for Solving 2-D Transient Coupled Heat Transfer in a Rectangular Semitransparent Medium," Numer. Heat Transfer, Part A, **55**(5), pp. 465–486.
[18] Chaabane, R., Askri, F., and Nasrallah, S. B., 2011, "Analysis of Two-Dimensional Transient Conduction-Radiation Problems in an Anisotropically Scattering Participating Enclosure Using the Lattice Boltzmann Method and the Control Volume Finite Element Method," Comput. Phys. Commun., **182**(7), pp. 1402–1413.
[19] Chaabane, R., Askri, F., and Nasrallah, S. B., 2011, "Parametric Study of Simultaneous Transient Conduction and Radiation in a Two-Dimensional Participating Medium," Commun. Nonlinear Sci. Numer. Simul., **16**(10), pp. 4006–4020.
[20] Zhao, J. M., and Liu, L. H., 2007, "Solution of Radiative Heat Transfer in Graded Index Media by Least Square Spectral Element Method," Int. J. Heat Mass Transfer, **50**(13–14), pp. 2634–2642.
[21] Canuto, C., Hussaini, M. Y., Quarteroni, A., and Zang, T. A., 2006, *Spectral Methods: Fundamentals in Single Domains*, Springer, Berlin.
[22] Boyd, J. P., 2001, *Chebyshev and Fourier Spectral Methods*, Dover Publications, New York.
[23] Peyret, R., 2002, *Spectral Methods for Incompressible Viscous Flow*, Springer, Berlin.
[24] Canuto, C., Hussaini, M. Y., Quarteroni, A., and Zang, T. A., 2007, *Spectral Methods: Evolution to Complex Geometries and Applications to Fluid Dynamics*, Springer, Berlin.

[25] Graham, N., Quandt, M., and Weigel, H., 2009, *Spectral Methods in Quantum Field Theory*, Springer-Verlag, Berlin.
[26] Shan, X. W., Montgomery, D., and Chen, H. D., 1991, "Nonlinear Magnetohydrodynamics by Galerkin-Method Computation," Phys. Rev. A, **44**(10), pp. 6800–6818.
[27] Orszag, S. A., 1974, "Fourier Series on Spheres," Mon. Weather Rev., **102**(1), pp. 56–75.
[28] Worcester, P. F., Cornuelle, B. D., Dzieciuch, M. A., Munk, W. H., Howe, B. M., Mercer, J. A., Spindel, R. C., Colosi, J. A., Metzger, K., Birdsall, T. G., and Baggeroer, A. B., 1999, "A Test of Basin-Scale Acoustic Thermometry Using a Large-Aperture Vertical Array at 3250-km Range in the Eastern North Pacific Ocean," J. Acoust. Soc. Am., **105**(6), pp. 3185–3201.
[29] Zenouzi, M., and Yener, Y., 1992, "Simultaneous Radiation and Natural Convection in Vertical Slots," Developments in Radiative Heat Transfer, HTD-Vol. **203**, pp. 179–186.
[30] Kuo, D. C., Morales, J. C., and Ball, K. S., 1999, "Combined Natural Convection and Volumetric Radiation in a Horizontal Annulus: Spectral and Finite Volume Predictions," ASME Trans. J. Heat Transfer, **121**(3), pp. 610–615.
[31] De Oliveira, J. V. P., Cardona, A. V., Vilhena, M. T., and Barros, R. C., 2002, "A Semi-Analytical Numerical Method for Time-Dependent Radiative Transfer Problems in Slab Geometry With Coherent Isotropic Scattering," J. Quant. Spectrosc. Radiat. Transf., **73**(1), pp. 55–62.
[32] Zhao, J. M., and Liu, L. H., 2007, "Spectral Element Approach for Coupled Radiative and Conductive Heat Transfer in Semitransparent Medium," ASME Trans. J. Heat Transfer, **129**(10), pp. 1417–1424.
[33] Li, B. W., Sun, Y. S., and Yu, Y., 2008, "Iterative and Direct Chebyshev Collocation Spectral Methods for One-Dimensional Radiative Heat Transfer," Int. J. Heat Mass Transfer, **51**(25–26), pp. 5887–5894.
[34] Sun, Y. S., and Li, B. W., 2009, "Chebyshev Collocation Spectral Method for One-Dimensional Radiative Heat Transfer in Graded Index," Int. J. Therm. Sci., **48**(4), pp. 691–698.
[35] Li, B. W., Tian, S. A., Sun, Y. S., and Hu, Z. M., 2010, "Schur-Decomposition for 3D Matrix Equations and Its Application in Solving Radiative Discrete Ordinates Equations Discretized by Chebyshev Collocation Spectral Method," J. Comput. Phys., **229**(4), pp. 1198–1212.
[36] Li, B. W., Sun, Y. S., and Zhang, D. W., 2009, "Chebyshev Collocation Spectral Methods for Coupled Radiation and Conduction in a Concentric Spherical Participating Medium," ASME Trans. J. Heat Transfer, **131**(6), p. 062701.
[37] Sun, Y.-S., and Li, B.-W., 2010, "Spectral Collocation Method for Transient Conduction-Radiation Heat Transfer," J. Thermophys. Heat Transfer, **24**(4), pp. 823–832.
[38] Chen, H. L., Su, Y. H., and Shizgal, B. D., 2000, "A Direct Spectral Collocation Poisson Solver in Polar and Cylindrical Coordinates," J. Comput. Phys., **160**(2), pp. 453–469.